

Computer Science 3 [MC2201]**Spring Semester, 2025**

Department	Mathematics and Computer Science	Credits	3	Instructor	Hyunwoo Jung	Class Room	Creation Building(CB) 702
Subject	Computer Science 3	Class Hours/Wk	3	E-mail			
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Office	Changjo 703						

Instructors would be available at any time after a prior appointment.

1. Course Description

Course Objectives	
This course is a general introduction to computer science. Students learn search engine, basic computer architecture, theory of computation, basic OS, networks, AI, data structures, and web programming. By learning these concepts in computer science, students possess general outline of subjects that occur in this field. This course provides you a basic milestone for further understanding of computer science that is essential in modern world.	
Contents	
1. Implementing PageRank Algorithm i. PageRank Algorithm ii. Implementation of PageRank 2. Basic Computer Architecture i. Data Representations ii. Toy Machine iii. Toy Instruction set iv. Toy Programming v. Basic Circuit Design 3. Algorithm i. Understanding Efficiency of an Algorithm ii. Divide and Conquer iii. Dynamic Programming	4. Threads and Processes i. Threads ii. Producer, Consumer, and Synchronization 5. Networks i. IP Addresses ii. Ports, Servers, and Clients iii. Sockets iv. Server Programming 6. AI I. Backpropagation Algorithm ii. Pytorch Programming 7. Simple Web App Programming i. Http Request, Http Response ii. MySQL, FastAPI, and React

2. Core Abilities

Course Core Abilities	
- Problem Solving, Inference, Communication, Attitude and Tenacity, Ethics	
- Problem Solving	Students learn basic concepts of CS. Students devise their diverse ideas to tackle problems
- Inference	Students develop themselves to understand abstract concepts and generalize them
- Communication	Students communicate with other peer students and the teacher through active participation and presentations
- Attitude and Tenacity	Students learn basic attitude to become scientist and importance of tenacity while learning fundamental concepts of CS
- Ethics	Students participate in the course works actively without dishonest actions. Students learn taking responsibility of dishonest actions.

3. Prerequisite

Basic programming skills that are acquired through computer science 1 and 2 courses

4. Course Info

No main textbook

Main materials can be downloaded from the course link:

Course Link

Google Classroom

5. Grading Policy

Individual Presentation	Homework Assignments	Midterm Exam	Attendance & Attitude
20%	20%	50%	10%

Individual presentation

A. We encourage students to present solution for homework problems and in-class problems.

Attendance & Attitude

A. Bad attitudes such as sleeping, chatting, etc. result in deduction.

6. Weekly Schedule

Weeks	Topics
Week1	PageRank Algorithm
Week2	Implementing PageRank Algorithm
Week3	Data Representation
Week4	Toy Programming
Week5	Circuit Design
Week6,7	Big O Notation

		Divide and Conquer
		Dynamic Programming
Week8		Midterm Exam
Week9		Threads and Processes
Week10		Networks, Server Programming
Week11,12		Basic AI
Week13		App Programming I
Week 14, 15		App Programming II
Week16		Final Exam Period

*** The schedule can be changed as it may chance.

7. Homework submission due date

There is a deadline for each assignment.

For a delay, a penalty is applied.

8. Policy for cheating

No cheating activity is allowed. Students who cheat, plagiarize, copy, fake, etc. will fail the course.

9. Proportion of A or B Grades

Basically, the proportion of students who get A or B grade is 70%. But if all students get high achievements, this ratio would be raised.

*The aim of this course is for you to have an overview of computer science. If you have any difficulty for taking this course, please feel free to contact the instructor.